

Strategic Fee Choice and Transaction-Priority Allocation under Incomplete Information

Lin Zhang

Duke Kunshan University
Kunshan, China

lin.zhang@dukekunshan.edu.cn

Zijie Wang

Duke Kunshan University
Kunshan, China

zijie.wang@dukekunshan.edu.cn

How Do Private Urgency and Waiting Shape Transaction Priority?

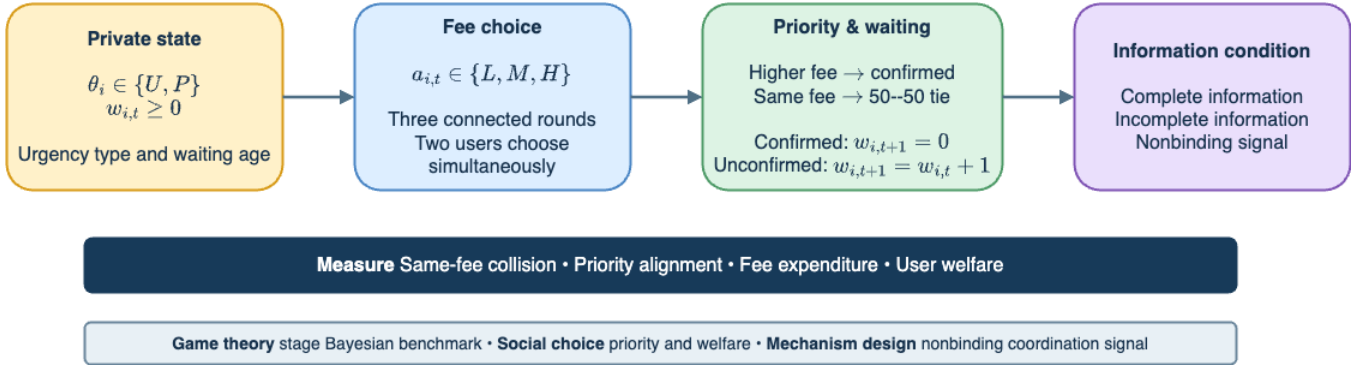


Figure 1: Research design. Two users with private Urgent or Patient types choose Low, Medium, or High fees across three connected rounds. Higher fees receive confirmation priority, while equal fees use a random tie-break and unconfirmed users accumulate waiting. We compare complete information, incomplete information, and incomplete information with a nonbinding coordination signal, evaluating priority allocation, fee expenditure, and user welfare.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** Team7. **Symposium:** September 28 Symposium.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Three connected research questions and verified gap

Blockchain systems must allocate limited transaction-processing capacity among users with different incentives, creating design questions for users, wallet designers, and protocol developers [4]. We study a simplified fee-choice setting motivated by Algorand [2]. Two users interact for three rounds and choose a Low (L), Medium (M), or High (H) fee. Each privately knows whether the transaction is Urgent or Patient. A higher fee receives confirmation priority; equal fees use a random tie-break. An unconfirmed user accumulates waiting, which changes incentives in later rounds.

Our shared question is how private urgency and accumulated waiting affect transaction-priority allocation, and whether a coordination signal can improve it. **Q1—Economics:** How do urgency, waiting, and fee costs shape strategic choices and welfare? **Q2—Computation:** How do information and coordination rules change

allocation outcomes? **Q3—Behavior:** How do users’ beliefs about the opponent affect fee choices and responses to recommendations?

Bayesian game theory provides a framework for strategic decisions under private information [3], while transaction-fee mechanism design studies allocation of scarce blockchain capacity [4]. Our project connects these questions in a dynamic setting where private types, discrete fee choices, and accumulated waiting jointly determine priority. Neither strategic prediction, computational allocation, nor behavioral choice alone resolves how the mechanism performs.

2 Answer to Q1: incentives and strategic model

There are two users $i \in \{1, 2\}$ with private types $\theta_i \in \{U, P\}$, where U is Urgent and P is Patient. In each round $t \in \{1, 2, 3\}$, both simultaneously choose

$$a_{i,t} \in \{L, M, H\},$$

with $c(L) = 1$, $c(M) = 2$, and $c(H) = 3$. We use $(V_U, d_U) = (8, 2)$ and $(V_P, d_P) = (5, 1)$, where V_θ is confirmation value and d_θ is delay cost.

The higher fee receives confirmation priority; equal fees use a 50–50 tie-break. If $w_{i,t}$ is waiting age before round t ,

$$w_{i,t+1} = \begin{cases} 0, & \text{if confirmed,} \\ w_{i,t} + 1, & \text{otherwise,} \end{cases} \quad u_{i,t} = \begin{cases} V_{\theta_i} - c(a_{i,t}), & \text{if confirmed,} \\ -d_{\theta_i} w_{i,t+1}, & \text{otherwise.} \end{cases}$$

The full setting is a finite-horizon dynamic Bayesian game. For a tractable game-theoretic prediction, we use a Bayesian Nash best-response benchmark at each simultaneous fee-choice stage [3]. Against H , choosing L or M gives $-d_\theta(w + 1)$, while

$$EU_\theta(H | H, w) = \frac{1}{2}[V_\theta - c(H)] - \frac{1}{2}d_\theta(w + 1).$$

Thus H is a strict best response to H when

$$V_\theta - c(H) + d_\theta(w + 1) > 0.$$

This holds for both types under our calibration, so (H, H) is a stage Bayesian Nash equilibrium. The prediction identifies the strategic tension: individually rational fee escalation can produce a same-fee collision instead of differentiated priority.

Game theory provides this strategic benchmark. Social choice evaluates whether scarce priority is aligned with urgency and the trade-off between allocation quality and user welfare. Mechanism design introduces differentiated, nonbinding recommendations and tests whether they improve allocation while choices remain voluntary.

3 Answer to Q2: computation, allocation comparison, and artifacts

The Colab implements the same three-round game. Its state-responsive comparison policy raises fees with urgency or waiting:

$$\pi(U, w) = \begin{cases} M, & w = 0, \\ H, & w \geq 1, \end{cases} \quad \pi(P, w) = \begin{cases} L, & w = 0, \\ M, & w = 1, \\ H, & w \geq 2. \end{cases}$$

This is a simulation policy rather than the equilibrium prediction above.

We compare the same priority-allocation environment under three information and coordination conditions. The complete-information benchmark coordinates differentiated recommendations using both users' types and waiting states. Under incomplete information, each user follows the comparison policy using only its own state. With a signal, the opponent's type remains private while the system provides nonbinding recommendations based on

$$q_{i,t} = 21\{\theta_i = U\} + w_{i,t},$$

with baseline compliance $\rho = 0.85$. The simulation samples types, applies the relevant policy or recommendation, allocates confirmation to the higher fee with random tie-breaking, and updates waiting for the next round.

The scarce resource is transaction confirmation priority. Fee choice determines priority, and the confirmed user's fee expenditure is the payment measure. We report same-fee collision, priority alignment, urgent-user confirmation, waiting, fee expenditure, individual utility, and total user welfare. In 100,000 simulated sessions, the signal reduces same-fee collision from 33.36% to 5.20% and raises priority alignment from 87.11% to 97.99%. Urgent-user confirmation rises from 66.39% to 81.25%, but mean fee expenditure rises from 2.352 to 2.812 and mean user welfare per round falls from 3.057 to 2.678. Thus the signal improves priority allocation at the cost of higher fee expenditure, rather than producing an automatic welfare improvement. Signal-compliance sensitivity confirms the same trade-off.

Computational scientist—GitHub:

https://github.com/z2005lin/PS2_Zijie_Lin

Runnable notebook:

<https://colab.research.google.com/drive/1E5NDNUU62rHPwp9UCG9oEHXplz3PG9EJ?usp=sharing>

Behavioral scientist—Hugging Face:

<https://huggingface.co/spaces/dku-comsci-econ206-2026/SubmitOrWait>

4 Answer to Q3: behavior and interdisciplinary integration

The rational benchmark predicts choosing H when a user expects the opponent to choose H . Actual choices may differ because of strategic uncertainty, bounded reasoning, or learning from previous outcomes [1]. The Hugging Face game implements the same three-round $L/M/H$ setting and records each player's prediction of the opponent's action, confidence, reasoning, and fee choice.

This design provides a discriminating test. Choices moving toward H when players expect H support the best-response benchmark; systematic responses to accumulated waiting support state-dependent behavior; and changes under the signal identify recommendation compliance. These observations can be compared with the computational benchmark and can reveal whether the comparison policy or signal interface needs revision. Classroom choices are treated as exploratory behavioral evidence rather than a population-level effect.

5 Advanced development, real-world impact, and roadmap

The project integrates strategic prediction, dynamic computation, and behavioral testing within one transaction-priority problem. Academically, it distinguishes individual best responses from allocation and welfare outcomes and identifies a trade-off between better priority coordination and higher fee expenditure. For industry, the comparison may inform wallet or transaction-priority interfaces; for users, it makes visible the cost of obtaining better priority alignment. If externally validated, nonbinding recommendations may provide a simple coordination tool when users differ in urgency, but improved allocation should not be interpreted as an automatic welfare gain.

The main risks are stylized payoffs, unequal effects across user types, and overextending a two-user model to a live blockchain market. We therefore report type-specific outcomes and treat Algorand as the motivating environment rather than empirical evidence. If behavioral evidence contradicts the benchmark, we will revise the comparison policy or signal design and rerun the computational tests. The next study will test participant beliefs and signal compliance under alternative fee menus and payoff calibrations.

Expected 2056 Nobel Prize and Turing Award contribution statement. We aspire to develop transparent and testable mechanisms for allocating scarce digital resources under private information.

Open Science Statement

The GitHub repository records the project files and implementation at review-ready commit 66afa2c, while the Colab reproduces the strategic benchmark, three-condition simulation, round-level results, and signal-compliance sensitivity analysis. The Hugging Face Space implements the corresponding behavioral game. Seeds, parameter definitions, and artifact versions are recorded so computational results can be reproduced.

Statement of Contribution to the UN's SDGs

The project relates to SDG 9 through the design of reliable digital infrastructure [5]. Its potential contribution is narrower than a demonstrated SDG impact: if validated, better priority-allocation interfaces may reduce coordination failures while preserving access for users with different urgency levels.

Author Notes

Acknowledgements

We thank Professor Luyao Zhang, course instructor for COM-SCI/ECON 206, for the course materials and guidance that shaped the development of this project. The PS2 annotated template, Week 4 strategic-game materials, and Week 5 auction and allocation materials were used to check the game structure, solution concept, and allocation interpretation.

The computational analysis was developed in Python and Google Colab, with GitHub used to maintain the project repository and computational outputs. Hugging Face Spaces hosts the behavioral implementation of the game, and the A0 poster was developed from the course poster template. OpenAI Codex and ChatGPT (GPT-5.6 Sol) were used as described in Appendix A.1 for coding, debugging, wording, LaTeX organization, and visual preparation. These tools did not replace the authors' decisions about the research question, model, assumptions, or interpretation. The authors remain responsible for the proposal, computational results, citations, and remaining errors.

Team and individual contributions

Lin Zhang and Zijie Wang jointly developed the central research idea and decided to study fee differentiation as a two-user strategic allocation problem under private urgency. Through discussion and revision, we agreed on the one-shot High/Low fee-choice setting, the comparison across information conditions, and the use of congestion, type sorting, and welfare as the main outcomes. Both authors reviewed the final model and its interpretation across the proposal, notebook, behavioral game, and poster.

Zijie Wang developed the initial proposal and the main computational implementation. He translated the shared research idea into the first formal model, implemented the Bayesian benchmark and simulation in Google Colab, and organized the GitHub repository containing the computational materials. His main inspectable outputs are the runnable notebook and the computational results used in Section 3 and the A0 poster.

Lin Zhang revised and integrated the proposal after the initial draft, with particular attention to the connection among the strategic model, computational results, behavioral design, and assignment requirements. She redesigned the Hugging Face Space to implement the same private-type High/Low fee-choice game and developed the A0 poster from the proposal and notebook results. She also revised the teaser figure and final wording so that the proposal, Colab, Hugging Face game, and poster use the same research question, assumptions, and evidence boundary. After symposium feedback, both authors will review any challenged assumptions and make the final revision jointly.

References

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A Technical details and reproducibility

The computational model implements the two-user one-shot fee-choice game defined in the main proposal. Each user has a private type $\theta_i \in \{U, P\}$ and chooses an action $a_i \in \{H, L\}$. Matching choices produce the congestion payoff (R_C, R_C) , while differentiated choices produce (R_H, R_L) or (R_L, R_H) , with $R_H > R_L > R_C$.

The baseline calibration uses

$$(R_H, R_L, R_C)_U = (8, 2, 1)$$

for Urgent users and

$$(R_H, R_L, R_C)_P = (5, 4, 1)$$

for Patient users. Under the baseline prior and payoff specification, the pure Bayesian benchmark is

$$U \rightarrow H, \quad P \rightarrow L.$$

The Colab notebook first computes type-dependent expected payoffs and verifies the pure Bayesian equilibrium by checking best responses over type-contingent strategies. It then compares three environments: a coordinated complete-information benchmark, decentralized choice under incomplete information, and incomplete information with a nonbinding coordination signal. The main outcomes are same-fee congestion, efficient type sorting, and total welfare. Additional sensitivity checks vary the prior probability of the Urgent type and the probability that an agent follows the coordination signal.

Under the baseline simulation, incomplete information produces substantially more same-fee congestion and lower welfare than the coordinated benchmark. Adding the nonbinding signal reduces much of this coordination loss. These outputs are conditional on the stated payoff calibration and simulated behavior. They are not participant data and are not presented as evidence of current Algorand network performance.

The runnable notebook is available through the project Google Colab, and the computational files and outputs are maintained in the GitHub repository. The review-ready repository version is Git commit 66afa2c. The Hugging Face Space implements the same strategic setting as an interactive behavioral task. It records the player's private type, belief about the other user's High-fee choice, confidence, reasoning, and final H/L decision. This behavioral interface is designed for a later comparison with the Bayesian benchmark; no population-level behavioral result is claimed in the current proposal.

A.1 AI-use disclosure

The central PS2 research idea was developed through discussion between the two authors.

OpenAI Codex was used during development to assist with coding and document implementation. Its main uses included transferring author-written material into the required LaTeX structure, debugging and organizing parts of the computational implementation, and refining the teaser figure.

ChatGPT using GPT-5.6 Sol was used during the later revision to improve wording, and organize author-written material into LaTeX. It was also used to help prepare selected visual materials for the A0 poster from the model and computational outputs. In these interactions, the authors supplied the research design, formulas, computational results, and intended interpretation, then reviewed and edited the resulting text and figures.

AI-generated suggestions were treated as drafting and implementation assistance. The authors checked the final strategic model, payoff definitions, equilibrium interpretation, computational outputs, citations, and evidence-status statements against the proposal and linked artifacts. Lin Zhang and Zijie Wang remain responsible for all final research decisions, claims, computations, and submitted materials.

B Cumulative Intellectual Development

The first PS2 version developed from a transaction-timing idea in which users chose whether to Submit or Wait under congestion. The early proposal and Colab therefore treated the problem as a dynamic interaction across rounds. This version helped identify the broader question that remained important to us: how users with different transaction urgency coordinate when access to transaction priority is scarce.

As we compared the model more closely with the PS2 strategic-game requirements, the dynamic Submit/Wait structure created unnecessary complications. Timing, changing states, and repeated decisions made it difficult to isolate the role of private information and also made the behavioral implementation harder to connect directly to a clean equilibrium benchmark. We therefore retained the transaction-priority problem but changed the strategic representation.

The current version uses a one-shot High/Low fee-choice game. Two users privately observe whether they are Urgent or Patient and choose simultaneously. Same-fee choices create congestion, while different fees produce differentiated priority. This revision made Bayesian equilibrium the natural incomplete-information benchmark and allowed the computational comparison to focus on complete information, incomplete information, and incomplete information with a nonbinding coordination signal.

The Colab analysis then made an additional issue visible. Under the baseline calibration, incomplete information produces much more same-fee congestion than the coordinated benchmark even though efficient type sorting remains high when differentiation succeeds. This shifted our emphasis from fee level alone to the coordination problem created by private information. The signal treatment was retained because it provides a simple way to test whether coordination can improve without revealing private types or forcing an action.

The Hugging Face game was subsequently redesigned around the same one-shot model rather than the earlier Submit/Wait setting. It asks players to observe their private type, report a belief and confidence level, and choose High or Low, with a later signal condition providing a nonbinding recommendation.

C Open-source and three-lens inspiration

The project uses Algorand as a motivating transaction environment rather than as an empirical dataset. Algorand’s design provides the broader setting in which users submit transactions and transaction priority is scarce, while the present model deliberately replaces the full protocol with two users and binary High/Low fee choices. The model therefore should not be read as a reconstruction of the live Algorand fee market.

The open computational workflow also shaped the project. Google Colab provides a runnable environment for the equilibrium checks, simulations, and sensitivity analysis, while GitHub records the code and outputs at a fixed version. Hugging Face Spaces provides a separate behavioral interface in which the same private-type decision can be experienced directly. These tools serve different purposes, but they use the same types, actions, payoff logic, and evidence boundary.

The three disciplinary lenses changed how we evaluated the same game. From a **game-theory** perspective, the main question is whether strategies are mutual best responses under complete or incomplete information. From a **social-choice** perspective, equilibrium alone is not enough; we also compare congestion, whether the higher-urgency user receives priority, and total welfare. From a **mechanism-design** perspective, the coordination signal is an intervention rather than a new objective. It asks whether differentiated recommendations can improve coordination while leaving private types private and choices voluntary.

D Review response and revision record

This appendix is reserved for the PS2 review and revision cycle. The current document is the September 27, 2026 first-draft version.

E Structured abstract and cumulative proposal map

Table 1 summarizes the current PS2 version and keeps the same eight-part structure used in the earlier proposal record. Computational results are separated from behavioral and live-network claims.

Table 1: Current PS2 structured abstract and cumulative proposal map.

Proposal element	Current statement / change
Background	Transaction priority is scarce, and users choosing fees without observing others' urgency face a coordination problem. Existing work provides foundations for Nash and Bayesian equilibrium, coordination, and transaction-fee design. The present project isolates a narrower question: whether two privately informed users can differentiate High and Low fee choices when matching choices create congestion.
Motivation and application	The motivating case is a simplified transaction-fee interface in which an Urgent and a Patient user may value priority differently. Failed coordination can place both users at the same fee level and produce congestion. The simplified setting allows the role of private information and coordination signals to be studied before introducing more users, continuous fees, or live-network dynamics.
Three connected questions	Economics asks how private urgency changes equilibrium, priority allocation, and welfare. Computation asks how the Bayesian benchmark and information treatments can be solved and compared reproducibly. Behavioral science asks whether actual users follow the type-dependent benchmark and whether a nonbinding signal changes their decisions. Together these questions ask when decentralized fee choices can successfully allocate scarce transaction priority under private information.
Method and model assumptions	Two users privately observe type $\theta_i \in \{U, P\}$ and simultaneously choose $a_i \in \{H, L\}$. Matching actions receive the congestion payoff, while differentiated actions assign High and Low rewards with $R_H > R_L > R_C$. The baseline calibration gives Urgent and Patient users different rewards and produces the pure Bayesian strategy $U \rightarrow H, P \rightarrow L$. The Colab compares a coordinated complete-information benchmark, incomplete information, and incomplete information with a nonbinding signal.
Results or expected results	The completed computational analysis verifies the baseline Bayesian equilibrium and shows a large coordination loss under incomplete information. In the baseline simulation, same-fee congestion is about 51.5% under incomplete information and about 12.6% with the coordination signal; mean total welfare increases from about 6.78 to 9.76. These are model-generated results. The planned behavioral test asks whether human choices and signal responses follow the same pattern.
Intellectual merits	The project separates three questions that are easy to combine incorrectly: what privately informed users should do under the model, whether decentralized choices produce an efficient allocation, and whether people actually follow the Bayesian benchmark. Linking equilibrium analysis, reproducible simulation, and a behavioral interface makes these distinctions directly testable within the same strategic game.
Practical impacts	If the mechanism is externally validated, the results may inform how transaction interfaces present discrete fee choices or nonbinding coordination recommendations. Possible benefits include lower congestion and better use of scarce priority, while risks include unequal access for Patient or lower-value users and oversimplification of a live network. The project therefore relates to SDG 9 through reliable digital infrastructure, but no practical impact is claimed from the current simulation alone.
Feedback received and revision made	The project initially used a repeated Submit/Wait congestion game. Internal review of the strategic model and assignment requirements led us to retain the transaction-priority question but replace the dynamic structure with a one-shot High/Low Bayesian anti-coordination game. The Colab, Hugging Face interface, teaser, and poster were then revised around the same types, actions, information conditions, and evidence boundary. Formal symposium feedback will be added after September 28.

F PS2 extension record: allocation, linked artifacts, poster, and symposium

F.1 Allocation record

The PS2 application is treated as an allocation problem rather than a continuous auction. The scarce resource is transaction priority. Each user privately observes an Urgent or Patient type and chooses High or Low. There is no continuous bid, monetary payment rule, or auction stopping rule in the current model.

The decentralized baseline allocates priority through simultaneous H/L choices under incomplete information. Successful differentiation gives one user High priority and the other Low priority, while matching choices produce congestion. The signal treatment keeps the same action set and private types but adds differentiated, nonbinding recommendations. The desired outcomes are lower same-fee congestion, priority sorting that reflects urgency, and higher total welfare.

This interpretation preserves the allocation lens required by the assignment without treating the binary fee-choice game as an auction that it does not implement. A richer transaction-fee mechanism with continuous bids or explicit payment rules would be a separate extension.

F.2 Linked-artifact parity

Artifact	Current status and connection
GitHub	Contains the computational project materials and outputs corresponding to the model in the proposal. The review-ready version is commit 66afa2c.
Google Colab	Implements the Bayesian benchmark, treatment simulations, and sensitivity checks used in the proposal and poster. Its outputs are model-generated rather than participant evidence.
Hugging Face	Implements the same private-type H/L game as a behavioral task. It records beliefs and choices and includes the nonbinding signal condition, but no population-level behavioral result is currently claimed.
A0 poster	Uses the same research question, baseline calibration, computational results, behavioral design, limitations, and open-material links as the proposal.

F.3 Symposium and practical-impact record

The A0 poster prepared for the September 28 symposium presents the baseline Bayesian benchmark, the three-condition computational comparison, the signal-compliance result, and the planned behavioral test. The main practical case is a transaction-fee interface that could provide simple, nonbinding coordination recommendations when users do not observe one another's urgency.

The potential benefit is reduced coordination failure and better use of scarce transaction priority. The main concern is that an intervention that improves aggregate welfare may still affect access

differently across user types. Before any practical recommendation, the payoff calibration would need external validation and the behavioral predictions would need to be tested with participants.

The September 28 presentation version, peer-evaluation forms, questions, and resulting Keep, Revise, or Investigate decisions will be retained for the final revision. Any change to the model or computational claim will be checked against the Colab and then updated consistently in the proposal, Hugging Face description, and poster.